Ayush Kumar

Software Engineer | Backend & Full Stack Developer

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Summary

Backend-focused Software Engineer with **2.5+ years** of experience owning and scaling production systems across e-commerce, loyalty, retail, payments, logistics and HR. Core expertise in **Node.js**, **TypeScript**, **REST APIs**, **microservices**, **SQL** and **Redis** on **AWS**, with experience designing high-throughput services supporting **100K+ customers**, **10,000+ concurrent users** and **10 lakh+ daily payout transactions**. Proven track record of designing **caching strategies**, **idempotent payment workflows** and **service decomposition** to improve **API** and **database performance** while modernizing legacy systems into microservices. Strong problem-solving foundation with **671+ LeetCode problems solved** and a **500 Days coding badge**.

Technical Skills

Languages	**JavaScript**, **TypeScript**, Java, Python, Go (GoLang)
Backend	**Node.js**, Express.js, **REST APIs**, **GraphQL**, **Microservices**, Spring Boot
Databases	**MySQL**, **MSSQL**, MongoDB, Oracle Database, **Redis**
Frontend	**React.js**, Angular, Vue.js, Flutter
Architecture	**Distributed Systems**, **System Design**, API Design, **Event-Driven Architecture**, **Caching**, Concurrency, Performance Optimization
Cloud & DevOps	**AWS**, **Docker**, Kubernetes, Jenkins, GitHub Actions, Nginx, Linux
Security	**JWT Authentication**, **RBAC**, OAuth, API Security, **OWASP Top 10**, VAPT Remediation, Input Validation, Encryption
Integrations	**Razorpay**, **PayU**, **Worldline**, **Shopify (REST/GraphQL)**, **SAP**, Freshdesk, Shiprocket, Delhivery, Firebase, APNs, Zillion
Tools	Git, Postman, VS Code, IntelliJ IDEA, Android Studio

Professional Experience

Software Engineer Mar 2024 – Present

DS Group *Noida, India*

- Owned end-to-end engineering delivery across **6+ production applications** spanning e-commerce, loyalty, payments, logistics and HR, taking features from technical design through deployment and production support.
- Scaled backend systems to support **100K+ customers** and **10,000+ concurrent users**, with production workloads processing **10L+ daily payout transactions** across payments and loyalty platforms.
- Drove the modernization of legacy **PHP** systems into independently deployable **microservices**, improving release velocity, fault isolation and long-term maintainability across the engineering org.
- Owned secure payment and payout workflows across **Razorpay**, **PayU** and **Worldline**, ensuring reliable transaction processing, reconciliation and fraud prevention at scale.
- Designed and maintained **250+ REST APIs** and third-party integrations spanning commerce, loyalty, payments and logistics workflows.
- Reduced deployment cycle time by **60%** by standardizing CI/CD and release workflows across **AWS**, **Docker**, **Jenkins**, PM2 and Nginx.
- Remediated **150+ high-severity VAPT findings**, raising the security posture of authentication, payment and API layers to **OWASP**-aligned standards.
- Led team-level project execution – breaking down technical work, unblocking teammates and reviewing solutions across multiple concurrent deliverables.
- Partnered directly with product, business and operations stakeholders to translate customer and logistics pain points into scalable engineering solutions, including an **AI-assisted logistics workflow** for OVINO aimed at reducing manual operational effort.

Key Projects

Rajnigandha Loyalty & CRM Platform Mar 2024 – Present

Node.js, React.js, TypeScript, MySQL, Microservices, Redis, Spring Boot, Razorpay, Shopify, SAP

- Designed and decomposed a legacy **PHP monolith** into independently deployable **Node.js microservices** (user management, coupon validation, reward distribution, payments), with clear service boundaries to enable independent scaling and deployment.
- Designed a multi-layer **caching strategy** using **Redis** combined with query indexing and connection pooling on **MySQL**, reducing query execution time by **65%** and improving high-frequency reads by **55%**.
- Engineered **idempotent** reward-redemption and payout workflows with **retry handling**, transaction validation and reconciliation logic to safely process **10L+ daily UPI**/bank-transfer payouts without duplicate disbursement.
- Designed scalable, stateless API layers to support **10,000+ concurrent users**, improving throughput and API performance by **40%+** through backend optimization and efficient data-access patterns.

- Built **Shopify REST/GraphQL** and **SAP** integrations for real-time order, invoice and customer synchronization, and extended CRM workflows with a **Freshdesk** integration that reduced support response time from **48 hours to under 6 hours**.
- Strengthened authentication and authorization using **JWT** and **RBAC**, and remediated **15+ VAPT findings** across API, payment and data-access layers.

Ksheer & OVINO Enterprise Dairy E-Commerce Jul 2024 – Present
Node.js, Angular, TypeScript, MSSQL, Razorpay, PayU, Worldline, Jenkins, Nginx

- Scaled the platform from **2,000+ to 50,000+ customers** while daily GMV increased from approximately **Rs. 20K to Rs. 3L+**, evolving the database and order-processing design to support growth.
- Optimized **MSSQL**-backed **Node.js** APIs through query optimization, indexing and efficient data-access patterns to maintain responsive order, customer and payment workflows under increasing transaction volume.
- Implemented recurring subscription and auto-pay workflows with **retry logic** and failure handling across **Razorpay**, **PayU** and **Worldline**, improving fault tolerance for recurring transactions and refund processing.
- Owned backend and platform engineering for customer-facing dairy e-commerce systems, covering product catalogue, checkout, order lifecycle, subscriptions, delivery scheduling, refunds and administrative workflows.
- Developed **Angular**-based administrative workflows for catalogue, orders, delivery operations and customer management, collaborating with product and operations teams to eliminate manual process bottlenecks.
- Implemented **JWT**-based authentication and **RBAC** across customer, delivery and admin workflows and resolved security issues identified through **VAPT** assessments.
- Built rollback-safe release workflows using Git, **Jenkins**, PM2 and Nginx, troubleshooting application and server issues to minimize production downtime.
- Contributed to an **AI-assisted logistics workflow** for OVINO to improve operational decision-making and reduce manual coordination across delivery processes.

DSphere – Employee Self-Service Platform Aug 2025 – Present
Node.js, React.js, MySQL, REST APIs, JWT, RBAC, PM2, Nginx

- Automated **200+ recurring HR requests per month** through self-service workflows, reducing dependency on manual HR operations and improving request resolution efficiency.
- Designed HR ticketing workflows with **SLA-based status tracking**, approval hierarchies and role-based access for employees, HR, administrators and super administrators.
- Built and maintained secure employee self-service workflows for HR documents, salary slips, Form 16, relieving letters, news feeds and HR ticketing, supporting self-service access to frequently requested information.
- Implemented **JWT authentication** and **RBAC** across sensitive employee-data APIs and handled production deployment, debugging and server configuration using Git, PM2 and Nginx.

Farm Connect – Field Operations Platform Dec 2024 – Present
Flutter, React.js, Node.js, Express.js, MySQL, Firebase, GoLang

- Developed a field-operations platform used by **500+ field agents**, combining **Flutter** mobile workflows with **React**-based administration and **Node.js** APIs for centralized data management.
- Implemented secure authentication and role-based access using **JWT/RBAC**, and an **event-driven** notification layer via **Firebase Cloud Messaging** combined with **GPS-based location tracking** for field activity monitoring.
- Optimized backend data access and API workflows and resolved **VAPT** findings across authentication and API layers before production deployment.

Education

GLA University 2020 – 2024
B.Tech. in Computer Science and Engineering | CGPA: 7.88/10 *Mathura, India*

Achievements

- **Star Performer – 2025-2026:** Recognized for engineering contribution, ownership and delivery impact at DS Group.
- **LeetCode:** Solved **671+ problems** across Data Structures, Algorithms, SQL and problem-solving; earned the **500 Days Badge** and multiple annual/daily challenge badges.

Certifications

- Machine Learning, Artificial Intelligence & Software Development Lifecycle – Udemy
- IoT / IIoT Communication – GLA University
- Data Science & Analytics – GLA University